

Low-LevelDesign(LLD)

Legal Document Customer Support RAG

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1 Module-Level Design

Document Processing Module

This module is responsible for loading the PDF file and extracting text content from it.

Chunking Module

The extracted text is divided into smaller overlapping chunks to improve retrieval performance and maintain context.

Embedding Module

This module converts text chunks into vector embeddings using a lightweight model (MiniLM).

Vector Storage Module

All embeddings are stored in a vector database (ChromaDB) for efficient similarity search.

Retrieval Module

This module retrieves the most relevant chunks based on the user query using similarity matching.

Query Processing Module

Combines the user query with retrieved context and prepares it for answer generation.

Graph Execution Module

Controls the overall workflow including routing, processing, and escalation logic.

HITL Module

Handles escalation logic.

2 Data Structures

Document Representation

```
{
  "content": "text",
  "metadata": {
    "page_number": 1
```

Chunk Format

```
{
  "chunk_text": "text content",
  "source": "document_id"
}
```

Embedding Structure

```
{
  "vector": [0.12, 0.45, 0.67, ...]
}
```

Query-Response Schema

```
{
  "query": "user question",
  "answer": "generated answer"
}
```

State Object(LangGraph)

```
{
  "query": "string",
  "route": "general/support/pricing",
  "answer": "string"
}
```

3 Work flow Design(LangGraph)

Nodes

- route_node → classifies query
- process_node → retrieves and answers
- hitl_node → handles escalation

Edges

- Route → process
- Route → hitl

State Flow

Query→RouteDecision→Processing/Escalation

4 Conditional RoutingLogic

Answer Generation Criteria

- Relevant chunks found
- Query is simple

Escalation Criteria

1. No relevant context
2. Answer is too short
3. Query is ambiguous

5 HITL Design

When Triggered

- Missing context
- Ambiguous question

Process

Query → Human Support → Response → Returned to User

6 API/Interface Design

Input

User query(string)

Output

Answer (string)

Flow

User → CLI → Processing → Output

7 Error Handling

Missing Data

Return: *"Not found in document"*

No Relevant Chunks

Fall back to HITL.

LLM Failure

Retry or escalate.

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